JAVA PRACTICAL PROGRAM REPORT

Bank Management System — All Test Cases

Program Name: BankManagementSystem.java

Language: Java

Starting Balance: 5000.0

This updated report includes the complete source code, algorithm, UML-style flow diagram, and output for each menu branch: check balance, deposit, successful withdrawal, insufficient balance, and invalid choice.

1. Aim

To demonstrate a menu-driven bank program and verify the output for every branch in the supplied code.

2. Algorithm

• Read the account holder name and display a welcome message.

• Display the three menu options and read the choice.

• For choice 1, display the current balance.

• For choice 2, read a deposit amount, add it to the balance, and display the result.

• For choice 3, read a withdrawal amount; withdraw only when amount is not greater than balance, otherwise show Insufficient Balance.

• For any other choice, show Invalid Choice.

• Close Scanner and end.

3. UML / Program Flow

BankManagementSystem balance = 5000
Read account holder name; display menu
Read menu choice
Choice 1: Check balance
Choice 2: Deposit and display new balance
Choice 3: Withdraw if sufficient; otherwise show error
Other choice: Invalid Choice; close Scanner

4. Complete Source Code

```
import java.util.Scanner;

public class BankManagementSystem {

    static double balance = 5000;

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("===== BANK MANAGEMENT SYSTEM =====");

        System.out.print("Enter Account Holder Name: ");
        String name = sc.nextLine();

        System.out.println("\nWelcome " + name);

        System.out.println("1. Check Balance");
        System.out.println("2. Deposit Money");
        System.out.println("3. Withdraw Money");

        System.out.print("Enter your choice: ");
        int choice = sc.nextInt();

        if (choice == 1) {
            System.out.println("Current Balance: " + balance);
        }

        else if (choice == 2) {
            System.out.print("Enter Deposit Amount: ");
            double amount = sc.nextDouble();

            balance = balance + amount;

            System.out.println("Money Deposited Successfully");
            System.out.println("Current Balance: " + balance);
        }

        else if (choice == 3) {
            System.out.print("Enter Withdraw Amount: ");
            double amount = sc.nextDouble();

            if (amount <= balance) {
                balance = balance - amount;

                System.out.println("Money Withdrawn Successfully");
                System.out.println("Current Balance: " + balance);
            }
            else {
                System.out.println("Insufficient Balance");
            }
        }

        else {
            System.out.println("Invalid Choice");
        }

        sc.close();
    }
}
```

5. Compilation and Execution

```
javac BankManagementSystem.java
java BankManagementSystem
```

6. Console Output — All Cases

Each case is run as a separate program execution, so the static balance resets to 5000.0 at the beginning of every case.

Case 1 — Check Balance

Shows the starting balance.

```
Enter Account Holder Name: Anu

Welcome Anu
1. Check Balance
2. Deposit Money
3. Withdraw Money
Enter your choice: 1
Current Balance: 5000.0
```

Case 2 — Deposit Money

Deposits 1500; balance becomes 6500.

```
Enter Account Holder Name: Anu

Welcome Anu
1. Check Balance
2. Deposit Money
3. Withdraw Money
Enter your choice: 2
Enter Deposit Amount: 1500
Money Deposited Successfully
Current Balance: 6500.0
```

Case 3 — Successful Withdrawal

Withdraws 1200; balance becomes 3800.

```
Enter Account Holder Name: Anu

Welcome Anu
1. Check Balance
2. Deposit Money
3. Withdraw Money
Enter your choice: 3
Enter Withdraw Amount: 1200
Money Withdrawn Successfully
Current Balance: 3800.0
```

Case 4 — Insufficient Balance

Withdrawal exceeds the starting balance; transaction is rejected.

```
Enter Account Holder Name: Anu

Welcome Anu
1. Check Balance
2. Deposit Money
3. Withdraw Money
Enter your choice: 3
Enter Withdraw Amount: 6000
Insufficient Balance
```

Case 5 — Invalid Choice

Choice 4 is outside the menu; program displays Invalid Choice.

```
Enter Account Holder Name: Anu

Welcome Anu
1. Check Balance
2. Deposit Money
3. Withdraw Money
Enter your choice: 4
Invalid Choice
```

7. Result and Notes

The five menu outcomes in the supplied code are covered. The code checks for insufficient balance during withdrawal, but it does not reject negative deposit or withdrawal amounts; these are additional validation cases not handled by the original program.